

Archimedean cross-term sharpening

PRIME.RDAGGER.CROSSTERM_SHARPENING.01 — round 476

Stefan Hamann

August 31, 2026

Abstract

The r474 diagnosis is implemented at fixed L : the slack was the archimedean cross-term, not the grid. The first Dirichlet H^2 ball with $\sigma_2 = (\pi/(2L))^2$ is one-dimensional, spanned by $h_0(x) = \cos(\pi x/(2L))$. Route B evaluates $Q_W(h_0)/\|h_0\|_2^2$ from the closed autocorrelation, the r475 regularized u -space pairing, a Bose prefix, and the exact algebraic tail $-L(\psi(K+1) - \psi(K + \frac{1}{4}))$. The Yoshida/Bombieri gate $L = 0.3$ (prime term identically zero) returns $r \in [1.163 \cdot 10^{-2}, 1.180 \cdot 10^{-2}] > 0$. Every sealed schedule point through $L_{16} = 2.7726$ is positive on this ball. This is not $\lambda_*(L) \geq 0$. Verdict UNCONDITIONAL($L_{\max} = 2.7726$) on the declared class. No RH claim.

Status. UNCONDITIONAL($L_{\max} = 2.7726$) on the first Dirichlet H^2 ball. No promotion of an RH claim.

Claim boundary. Nothing here is $\lambda_*(L) \geq 0$ on the r461 L^2 class. The theorem is the Rayleigh number of a single smooth generator. The archimedean symbol $\sigma_A(0) < 0$, so A is not a positive operator; Π supplies the reserve at low frequency. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.

1 Declared class

After r474, the working comparison class is

$$\mathcal{B}(L) = \{h \in H_0^2([-L, L]) : \|h''\|_2 \leq (\pi/(2L))^2 \|h\|_2\}.$$

The first Dirichlet eigenvalue of $-d^2/dx^2$ on $[-L, L]$ with vanishing endpoints is $(\pi/(2L))^2$. Hence $\mathcal{B}(L) = \text{span}\{h_0\}$, $h_0(x) = \cos(\pi x/(2L))$. No silent L^2 enlargement.

2 Closed data

Write $\omega = \pi/(2L)$. The autocorrelation $g = h_0 * \tilde{h}_0$ is even, supported in $[-2L, 2L]$, and

$$g(t) = \left(L - \frac{t}{2}\right) \cos(\omega t) + \frac{\sin(\omega t)}{2\omega} \quad (t \in [0, 2L]),$$

with $g(0) = \|h_0\|_2^2 = L$. The r471 form is $Q_W = A - P + \Pi$, scramble-sensitive: $P(g) = \sum_{n \leq e^{2L}} 2\Lambda(n)n^{-1/2}g(\log n)$. At $L = 0.3$ one has $2L < \log 2$, so $P \equiv 0$.

Lemma 1 (Regularized arch pairing). *With $w(u) = 2e^{-u/2}/(1 - e^{-2u})$ and $C_b = -(\gamma + \log \pi) - \log(1 - e^{-4L})$,*

$$A(g) = C_b g(0) + \int_0^{2L} w(u)(e^{-3u/2}g(0) - g(u)) du.$$

The integrand is regular at $u = 0$. Expanding $w(u) = 2 \sum_{k \geq 0} e^{-(2k+1/2)u}$ and integrating termwise against the closed g yields a Bose prefix of $K = 200$ terms plus the exact algebraic tail $-L(\psi(K+1) - \psi(K + \frac{1}{4}))$ and an $O(K^{-2})$ cubic remainder.

Lemma 2 (Pole, closed). $\Pi(g) = 2(I(-\frac{1}{2}) + I(\frac{1}{2}))$ with $I(\alpha) = \int_0^{2L} e^{-\alpha u} g(u) du$ elementary.

Lemma 3 (Symbol at zero). *The Fourier symbol of A obeys $\sigma_A(0) = -(\gamma + \log \pi) - (\pi/2 + 3 \log 2) \in [-5.37219, -5.37218]$. In particular A is negative at zero frequency. The series $\sigma_A(t) = -(\gamma + \log \pi) + \sum_k 2[(2k+2)^{-1} - \alpha_k/(\alpha_k^2 + t^2)]$ meets the closed value after the same ψ -tail.*

Lemma 4 (Peano constant of the regularized kernel). $C_0 := \int_0^\infty w(s) \frac{s^2}{2} ds = \sum_{k \geq 0} 2\alpha_k^{-3}$ with $\alpha_k = 2k + \frac{1}{2}$, and $C_0 \in [16.16596, 16.16597]$. Against an even increment with $|\Delta g(s) - \Delta g(0)| \leq (s^2/2) \|\Delta g''\|_\infty$ this is the L^1 factor of the regularized pairing (PROVED constant; not used for the Rayleigh table).

3 Theorem

Theorem 1 (First-mode unconditional). *On $\mathcal{B}(L)$ one has $Q_W(h)/\|h\|_2^2 = r(L)$ with $r(L)$ the sealed interval of Section 4. In particular $r(L) > 0$ at every schedule point, including the Yoshida/Bombieri value $L = 0.3$ and $L_{16} = 4 \log 2 = 2.7726$.*

The combination implication of r474 is not needed: there is no interpolant and no $\omega_L(\delta)$ on this one-dimensional ball.

4 Sealed balance

Probe `crossterm_probe.py`, SPEC_SHA `a123650e757b912d`. Smoke 15/15 in 0.14s; full 25/25 in 0.79s. Every row is H2BALL_POS. YB-gate PASS.

L	r_{lo}	r_{hi}	$\#\{p^k\}$	verdict
0.30 (YB)	$1.163 \cdot 10^{-2}$	$1.180 \cdot 10^{-2}$	0	H2BALL_POS
0.80	$1.837 \cdot 10^{-4}$	$2.080 \cdot 10^{-4}$	3	H2BALL_POS
$L_5 = 0.866$	$9.209 \cdot 10^{-4}$	$9.416 \cdot 10^{-4}$	4	H2BALL_POS
1.0	$1.099 \cdot 10^{-4}$	$1.254 \cdot 10^{-4}$	5	H2BALL_POS
1.2	$8.729 \cdot 10^{-5}$	$9.808 \cdot 10^{-5}$	8	H2BALL_POS
$L_8 = 1.386$	$1.389 \cdot 10^{-4}$	$1.470 \cdot 10^{-4}$	10	H2BALL_POS
1.4	$8.724 \cdot 10^{-5}$	$9.517 \cdot 10^{-5}$	10	H2BALL_POS
1.6	$9.875 \cdot 10^{-5}$	$1.048 \cdot 10^{-4}$	13	H2BALL_POS
1.8	$9.679 \cdot 10^{-5}$	$1.016 \cdot 10^{-4}$	18	H2BALL_POS
2.0	$7.191 \cdot 10^{-5}$	$7.579 \cdot 10^{-5}$	24	H2BALL_POS
$L_{12} = 2.079$	$2.456 \cdot 10^{-5}$	$2.816 \cdot 10^{-5}$	27	H2BALL_POS
$L_{16} = 2.773$	$3.452 \cdot 10^{-6}$	$5.473 \cdot 10^{-6}$	70	H2BALL_POS

At $L = 0.3$, $A \approx -0.28954$ and $\Pi \approx +0.29305$. The pole reserve beats the negative well of σ_A . The L_{16} margin is thin but strictly positive.

5 What this does not do

It does not prove $\lambda_*(L) \geq 0$ at any L . It does not restate a cofinal quantifier. High modes and functions that fail $h(\pm L) = 0$ remain outside $\mathcal{B}(L)$. No RH claim.